

REWEAR-FUSED: The Definitive Demo-Day & Judge-Defense Playbook

Cox Enterprises "Play With Purpose" Sustainability Hackathon · Carter's "Making & Remaking" Track · Atlanta · 17 June 2026

TL;DR

- **Win condition:** Treat this 72-hour build as a deep-tech investor pitch, not a coding contest. Lead with a one-killer-stat hook, run a fully offline deterministic "demo mode" with a 90-second pre-recorded MP4 as the silent fallback, and declare your TRL honestly (computational designs at TRL 2-3 with benchmark-referenced confidence scores) — calibrated honesty wins deep-tech rooms; over-claiming loses them (a16z Bio, "16 Pitfalls to Avoid When Building a Computational Therapeutics Company," 30 April 2018).
 - **Pitch the loop, not the molecule:** a single protagonist (a 4-year-old's stretch onesie), a single villain (the 2-5% elastane that "poisons" the entire recycle stream — Wrenby's/TU Wien, 2024), and a single bookended closing line that lands on a measurable Carter's-aligned KPI. The U.S. EPA's 2018 textile data (the most recent dataset) reports **11.3 million tons of textile MSW landfilled** in the U.S. that year (<https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/textiles-material-specific-data>); Carter's holds **19% of the U.S. baby apparel market (ages 0-2)** per Statista citing Carter's investor data (2022).
(Statista)
 - **Defend with structure, not bravado:** pre-empt the eleven predictable hostile questions (TRL, validation, cost, customer, moat, toxicity, scale, novelty, etc.) by **building the answer into the demo UI itself** — visible confidence scores, a "what we'd validate next" panel, predicted-byproducts safety panel — and use the **Bridge-Hook-Flag** structure (Grossman Group) so every "I don't know" becomes "Here's what we know, here's what we'd test next, here's why that's the right next dollar."
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Key Findings

1. **The single highest-leverage investment is rehearsal volume against a hostile mock panel**, not slide polish. Techstars founders practiced daily for the final three weeks; Keen.io's Ryan Spahn practiced "for at least four hours per day outloud — basically until my voice hurt" the final week ([keen.io/blog](https://www.keen.io/blog)). The murder-board / red-team technique — formalized at NASA by Hans Mark in the 1970s and used today by Supreme Court nominees and YC founders alike (Wikipedia: Murder board) — is the single most predictive activity for Q&A survival. (Keen) (Wikipedia)

2. **A 3-to-5-minute deep-tech pitch has a fixed beat structure:** 15-second hook → 30-second problem with one killer stat → 45-second solution + science-made-legible → 60-90-second live demo → 30-second business case/TAM → 15-second team → 15-second ask + closing line. This mirrors the YC Demo Day format (~2.5 minutes per pitch, Quartz, March 2016) and the 5-minute pitch + 7-minute Q&A used by CU Boulder's Deep Tech New Venture Challenge.
3. **In-silico deep-tech projects WIN by naming their TRL out loud and showing benchmarks,** LOSE by hiding behind jargon. NASA's TRL 1-9 scale (Mankins 1995; ISO 16290:2013) is the universal investor language; TRL 4-7 is the "valley of death" (Vendor.energy, 2025). a16z Bio (Low, Pande, Conde, 30 April 2018) is the canonical citation: *"For a company to create useful drugs, its in silico predictions — predictions made by computers — must eventually be proven out in biology, too... Wet lab work (i.e., biochemical or cellular experiments) must therefore validate the technology's biological predictions."* Investors assume you're over-claiming unless you say this first. (a16z)
4. **Demos fail predictably, and every failure is preventable.** The canonical playbook (Peter Cohan, *Great Demo*; Steve Jobs's 2007 Macworld iPhone reveal, where multiple backup iPhones were staged): pre-recorded MP4, local-only hosting, airplane mode, second laptop, deterministic "demo mode," extras of every cable. The single most common cause of avoidable failure is dependence on event Wi-Fi (AngelHack 2026 hackathon guide).
5. **Sustainability judging in 2026 is no longer vibes-based.** Cox Enterprises has now invested **>\$3 billion in cleantech since 2007** (Cox Enterprises press release, 21 May 2026); the Cox Cleantech Accelerator (a partnership with gener8tor and the Georgia Cleantech Innovation Hub) invests **\$250,000 per company** in its 12-week accelerator (gener8tor.com; Cox press release, 26 March 2026), and the separate equity-free **Cox Cleantech Residency** launched 14 May 2026 with a \$10,000 stipend for pre-seed Southeast founders. These judges have seen real LCAs and will detect greenwashing instantly. Speak in **\$ per ton CO₂ avoided, Scope 4 (avoided) emissions, and Additionality** (WePitched, 2026), and frame REWEAR-FUSED against named regulatory tailwinds — specifically the EU Digital Product Passport for textiles, with **textile-specific delegated acts expected to be adopted in 2027 followed by a minimum 18-month transition period, implementation starting from 2028 onward** (Carbonfact, citing EC ESPR Working Plan COM(2025)187 final; Intertek, 28 May 2025). (The Auto Channel + 8)
6. **Carter's-specific framing wins this track.** Carter's relaunched **KIDCYCLE™** in select stores in 2025 (with TerraCycle, originally launched 2021 — Carter's investor relations / DCFmodeling.com); **99% of Carter's apparel meets OEKO-TEX® STANDARD 100** in 2025 (DCFmodeling.com); Carter's holds **19% of the U.S. baby apparel market for ages 0-2** (Statista citing Carter's investor data, 2022). The "Making & Remaking" track is explicitly about circularity for stretch apparel where elastane currently breaks every recycling stream. Tie every line to a Carter's-internal metric. (DCFmodeling.com + 3)
7. **The single most consistent loser pattern at hackathons is "great tech, no business case, demo broke, ignored sponsor criteria, buried the lede"** ((code42cate) on DEV; BizThon on

Medium; AngelHack 2026). Corollary winner pattern: clear problem framing, audience-tailored language, one killer demo moment, memorable close.

Details

PART 1 — THE LIVE DEMO: MECHANICS, FAILURE MODES, AND BULLETPROOFING

1A. The Bulletproof Live-Demo Playbook

The "demo mode" deterministic walkthrough. Build a single hard-coded path through the product — a scripted user journey with fixed inputs, cached outputs, and no branching. I&CO's hackathon guide (Medium) names this directly: *"Having a team member write out a user journey that will act as the product demo is a good way to keep structure around your idea and prevent it from ballooning out of control."* For REWEAR-FUSED specifically, the deterministic demo path is: (1) user uploads a Carter's stretch onesie SKU's bill-of-materials JSON → (2) Digital Recyclability Passport renders with a red "NON-CIRCULAR — 4% elastane" flag → (3) one click switches to REWEAR-FUSED's cleavable elastane variant → (4) the 3D molecular viewer animates the carbamate-hydrolase enzyme docking and cleaving the new fiber → (5) the passport flips to green "98% MASS-RECOVERABLE" with a confidence interval. Every value, including the "live" molecular animation, is pre-computed and replayed from disk — there is no inference at demo time. (Medium)

The pre-recorded backup video. Industry consensus from Matt Boardman (Startup Elevator Pitch Coach), Atlassian/Loom (2020), and Srinathmohan (Medium, 2024): record a 90-second 1080p MP4 (1920×1080, H.264, ≤50 Mbps so it plays from USB stick on any laptop) of the full demo path, narrated, with captions burned in (Advids reports subtitle-optimized investor videos see "28% higher engagement because most investors view content on mute"). Save it in **three places**: the demo laptop's desktop, a USB-C stick taped to the back of the laptop, and the backup laptop. The single pivot line, rehearsed verbatim: *"The live build is running on a local server — let me show you the same flow from this morning's recording so we can talk through it."* No apology. No fumbling. Move on in <8 seconds. (Advids)

Local-vs-cloud hosting / avoiding live network dependencies. Run the entire web app on `localhost` on the demo laptop. Use a static build (Next.js `next export`, Vite production build, or a pre-rendered HTML/JS bundle) served by a local Python `http.server` or `npm serve`. Mock every API call with a local JSON fixture file — no fetch to the open internet. Put the laptop in **airplane mode the moment you walk into the venue**, and only re-enable Wi-Fi after the pitch. The AngelHack 2026 guide flags "WiFi failure" as the #1 organizer/participant headache; the

MLH API-demo guide recommends a personal hotspot as failover, but the better answer is **don't depend on Wi-Fi at all**.

The second-laptop principle. From Steve Jobs's 2007 iPhone unveiling: multiple backup iPhones were hidden on stage. Your equivalent: a second laptop, fully mirrored (same OS user account, same app, same demo mode, same backup MP4 on disk), already cabled to the projector adapter, already in airplane mode, already on the demo screen, sleep disabled. If laptop A freezes, you unplug HDMI from A, plug into B, and continue in <15 seconds. Everclear Marketing's Investor Demo Day guide: *"Bring extras of all equipment — laptops, power cords, USB drives, phones, whatever your demo requires."* (Matt Boardman) (Everclearmarketing)

Screen mirroring / projector compatibility. Atlanta convention venues typically present **HDMI** (most common), occasionally **VGA**, and increasingly **USB-C / DisplayPort over USB-C**. Bring a USB-C → HDMI adapter, a USB-C → VGA adapter, a USB-C → USB-C-DP cable, **and** a Mini-DisplayPort dongle for older MacBooks. Target output resolution **1920×1080 @ 60 Hz** (universally supported); set your laptop display to match so there is no scaling surprise. Set browser zoom to **150%** so the back of a 50-person booth area can read it. MLH's best-practice guide: *"I highly recommend asking the event organizers for an AV check before going onto the stage. Make sure you also check your laptop before presenting."* (Mlh)

Browser / display hygiene checklist (run 60 minutes before pitch):

- Fresh Chrome/Firefox profile, no extensions, no autofill, no bookmarks bar, no open tabs except the one demo URL.
- Full-screen mode (F11). Hide bookmarks bar, dock, taskbar, menu bar.
- Do Not Disturb / Focus mode ON. macOS: System Settings → Focus → DND. Windows: Settings → System → Notifications → Off. Disable Slack, Discord, iMessage, email.
- Energy: power adapter plugged in. Sleep: 1 hour minimum. Screen-saver off. Display-off-on-AC: never.
- Cursor visible and large (Accessibility → Pointer size 2× normal).
- All system fonts loaded locally — no (@import) from Google Fonts. Self-host any custom font.
- Close every other app. RAM-only the browser, the slide deck, and a Finder/Explorer window.
- Battery on backup laptop: ≥90%. Charged power bank in pocket.

1B. Every Common Demo-Failure Mode (and the Specific Counter)

Failure mode	Specific counter
Wi-Fi fails / slow	Run entirely on localhost. Personal mobile hotspot as backup.
Live API fails	All API responses cached as local JSON fixtures; "demo mode" toggle bypasses live calls.
GPU/compute too slow	Pre-render all 3D frames as a 60 fps MP4; the "live" Three.js scene is decorative only.
Laptop sleeps / notifications pop	DND on; sleep disabled; close email/Slack; remove sticky notifications.
Projector resolution mismatch	Set laptop to 1920×1080 @ 60 Hz before plug-in. Bring 4 adapters.
Fonts don't load	Self-host all fonts in <code>/public/fonts</code> ; no Google Fonts CDN.
WebGL crashes / browser memory leak	Pre-recorded flythrough MP4 as fallback. Restart browser between rehearsal and live.
Dead battery / power outlet missing	Power adapter cabled; 20,000 mAh power bank; charged backup laptop.
Dongle incompatible	Carry USB-C→HDMI, USB-C→VGA, Mini-DP→HDMI, and a 2 m HDMI cable.
Timing overruns	Rehearse to 95% of the time budget; teammate timekeeper holds a finger-count from the side.
Live demo crashes mid-pitch	One sentence: <i>"Let me show you the recording from this morning."</i> Switch to MP4. <8 seconds.

1C. 3D / WebGL / Molecular Rendering Specifics

REWEAR-FUSED's molecular viewer should use **3Dmol.js** or **Mol*** — both WebGL-based, peer-reviewed (Rego & Koes, *Bioinformatics* 31:8, 2015), production-grade, and used by RCSB PDB, UniProt, EMDB, and AlphaFold DB. Do **not** roll your own Three.js molecular renderer in 72 hours. Per ThreeJSResources.com (2025): WebGL 2.0 is supported in all modern browsers but performance "varies significantly across mobile devices" — assume a mid-tier venue laptop GPU.

[ResearchGate](#) [Three.js Resources](#)

Performance preflight checklist:

- Cap the scene to $\leq 15,000$ atoms (the original 3Dmol.js benchmark scene is 12,375 atoms and ran on a 2008 MacBook). (PubMed Central)
- Lock framerate to 30 fps if needed ((`renderer.setAnimationLoop`) throttled).
- Disable shadows, ambient occlusion, anti-aliasing on draw if FPS drops below 24.
- Pre-load the PDB/MOL file as a local file (no remote fetch).
- Profile in Chrome DevTools → Performance with 4× CPU slowdown to simulate a stressed conference laptop.

Fallback hierarchy (auto-cascade on failure):

1. **Live interactive WebGL** (best).
2. **Static high-res renders:** 8 pre-rendered PNG snapshots of key moments (enzyme docking, cleavage transition state, products) at 4K, clickable carousel.
3. **Recorded flythrough:** a 30-second MP4 of the camera circling the enzyme-substrate complex, annotated.
4. **The 90-second narrated demo MP4** (the master fallback).

1D. Psychology, Staging, and Physical Artifacts

The science-communication literature is unambiguous: **show, don't tell**. Randy Olson — marine-biology-PhD-turned-filmmaker, author of *Don't Be Such a Scientist* (Island Press, 2009; 2nd ed. 2018) and *Houston, We Have a Narrative* (2015) — argues scientists must aim "below the neck but above the belt" (heart and gut, not just head) and use the **ABT framework (And, But, Therefore)** as the spine of every explanation. For REWEAR-FUSED: *"Children outgrow clothes every 4 months **AND** Carter's KIDCYCLE keeps tens of thousands out of landfill, **BUT** every stretch garment with 2% elastane still poisons the recycling stream, **THEREFORE** we redesigned both the fiber and the enzyme that breaks it."* (Mountain Beltway)

Physical props to bring to the booth (cost <\$30 total):

- A Carter's stretch onesie — real, scuffed, with a visible "95% cotton, 5% spandex" tag. Hand it to the judge.
- A small petri dish or 50 mL Falcon tube containing dyed-water "enzyme buffer" with a fiber sample — clearly labeled as a visual aid, not a real reaction.
- A printed QR code on heavy cardstock, $\geq 8 \times 8$ cm, contrast ratio $> 7:1$, linking to a static landing page with deck, video, and contact info. Test the code with three different phones at booth-distance lighting before the event.
- A single-page printed leave-behind: problem, solution, one chart, team, contact. 100 copies (Everclear Marketing's booth-prep guide: *"Bring more than you think you may need"*).

(Everclearmarketing)

"On-screen vs. say-aloud" principle. Slides carry the **proof** (numbers, molecular structure, LCA delta). You speak the **story** and the **stakes**. Never read your slides — the YC Series A pitch guide is explicit: *"If you're trying to make more than one point per slide, your slide is too complicated."*

Y Combinator

PART 2 — PITCH NARRATIVE & STRUCTURE FOR A MIXED PANEL

2A. The Optimal 4-Minute Deep-Tech Beat Sheet (timed)

Time	Beat	Content
0:00– 0:15	Hook	One sentence + one image. <i>"This is a Carter's 4T stretch onesie. Right now, 100% of these go to landfill or downcycling — because of 4% of one fiber."</i> (Hold up the onesie.)
0:15– 0:45	Problem + Killer Stat	EPA 2018: 11.3 million tons of textile MSW landfilled (7.7% of all U.S. MSW landfilled). (US EPA) Carter's holds 19% of U.S. baby apparel (ages 0–2) (Statista) per Statista/Carter's investor data, 2022. Every blended-elastane garment "poisons" the recycling stream.
0:45– 1:30	Solution: the loop	Three components on one slide: A (cleavable elastane fiber), B (matched de novo carbamate-hydrolase enzyme), C (Digital Recyclability Passport software). One visual: a circular diagram.
1:30– 2:45	Live demo (75s)	Deterministic walkthrough: upload → red flag → swap to REWEAR-FUSED fiber → enzyme animation → green pass → confidence scores shown.
2:45– 3:15	Science made legible + TRL honesty	<i>"These are in-silico designs at TRL 2–3. Our carbamate-hydrolase scores a pLDDT of [YOUR VALUE] on AlphaFold and a Rosetta interface $\Delta\Delta G$ that beats the published UMG-SP-2 urethanase baseline. The next milestone is wet-lab expression at a partner lab in Q4."</i>
3:15– 3:45	Business case	Carter's-aligned: ties into KIDCYCLE (relaunched 2025); supports EU Digital Product Passport compliance — delegated acts adopted in 2027, minimum 18-month transition, implementation from 2028 onward (Intertek) (Intertek, 28 May 2025; Carbonfact citing EC ESPR Working Plan COM(2025)187 final). Statista values the U.S. children's apparel market at \$54.62B in 2025 (Statista) — name the wedge segment (US blended-stretch baby apparel) and bottom-up size it. Cox Cleantech Residency as commercialization pathway.
3:45– 4:00	Team + Ask + Close	One sentence each. Closing line bookends the hook: <i>"A 4-year-old will outgrow this onesie in 90 days. We want the next one to outlive the kid in a closed loop — and we want to do it from Atlanta."</i>

This mirrors and adapts the canonical YC anatomy: elevator pitch / problem / "we don't suck" / team / "we're fast" / ask (Bryce Roberts, OATV, on YC Demo Day; Geoff Ralston, *Guide to Demo Day Presentations*, Y Combinator blog).

2B. Making Complex Science Legible Without Dumbing It Down — Layered Messaging

The "explain to a smart 12-year-old then add one layer" technique works because of what Olson calls the **Four Organs** model — head, heart, gut — and it is mechanically executable on every slide via **paired language**: (Mountain Beltway)

Lay layer (heart/gut)	Technical layer (head)
"A scissor enzyme that only cuts elastane"	"De novo carbamate hydrolase, RFdiffusion-designed, scaffolded on a UMG-SP-2 motif"
"We redesigned the rubber-band thread in your kid's leggings"	"Enzymatically-cleavable poly(carbamate-urethane) with engineered weak bonds at every Nth repeat unit"
"We grade the garment like a credit score"	"Digital Recyclability Passport: probabilistic LCA + GHG ledger + SVHC chemical-compliance manifest"
"Confidence: 8 out of 10"	"pLDDT [YOUR VALUE], Rosetta $\Delta\Delta G$ [YOUR VALUE]"

Layered messaging is **simultaneous**, not sequential. Both audiences hear both layers; the business judge gets the metaphor; the technical judge gets the metric. KPMG's UK pitch guide calls this "**The Granny Test**" — your grandmother must understand what you do, and you must still be able to defend it to a domain expert. (KPMG)

2C. The TRL / Aspirational Framing — Calibrated Honesty Wins

The single biggest trap in deep-tech is **overclaiming on validation**. a16z Bio's Vijay Pande & Jorge Conde, April 30, 2018 (<https://a16z.com/16-pitfalls-to-avoid-when-building-a-computational-therapeutics-company/>): *"For a company to create useful drugs, its in silico predictions — predictions made by computers — must eventually be proven out in biology, too... Wet lab work (i.e., biochemical or cellular experiments) must therefore validate the technology's biological predictions and provide the insight that helps answer 'how it works' — especially for black box AI techniques."* (a16z)

For REWEAR-FUSED, declare TRL on a labeled slide. Use the NASA 9-point scale (Mankins, 1995; ISO 16290:2013; EU Horizon Europe). State out loud:

"REWEAR-FUSED is **TRL 2-3**: validated in-silico with benchmark-referenced confidence. The fiber is designed against a published depolymerization-rate model; the enzyme is designed against the UMG-SP-2 urethanase family (Magalhães et al., drain-metagenome paper, PMC12421990, 2025) and scored by AlphaFold pLDDT and a Rosetta $\Delta\Delta G$ benchmark. We have not yet wet-lab-validated. Our 90-day plan is to express the top-3 enzyme designs at

| a partner academic lab and run a Humicola insolens cutinase comparator (Cutinase reference: PMC8838978)."

This sentence kills 80% of hostile Q&A in advance. Investors **prefer** founders who say "TRL 3 and here's our wet-lab plan" over founders who say "we've built it." Equidam (2025) and Lindeninnovation's IRL guide (2024) confirm TRLs as standardized investor vocabulary; KTH's Innovation Readiness Level adds Customer/Market readiness as a parallel axis — name both.

2D. Storytelling Mechanics

Single protagonist, single villain. Per Pixar's "story spine" (Kenn Adams / Emma Coats, 2012): the protagonist is the 4-year-old in the onesie; the villain is the 4% elastane; the inciting incident is the moment the parent drops it in the KIDCYCLE bag and it gets sorted into the downcycle pile because of the elastane. The Pixar 6-step spine ("Once upon a time... every day... until one day... because of that... until finally... ever since then...") maps directly to a 4-minute pitch.

Myframework

Automateed

Emotional resonance. A child's garment in your hand is a more powerful prop than a landfill stat. Hold it.

Bookend/callback. The closing line must explicitly callback the opening image. The onesie is on stage at second 0 and at second 240.

The "one killer stat" principle. Reject five stats; choose one. The candidate: "**11.3 million tons of textile waste went to U.S. landfills in 2018 (EPA) — and Carter's ships one out of every five baby garments in America. We are the only team here designing both the fiber and the enzyme that closes that loop.**"

2E. Data Density Principles

YC Michael Seibel: "*Calculate market size with transparent math — show your work.*" Use bottom-up market sizing only. For REWEAR-FUSED: anchor on Statista's **\$54.62B U.S. children's apparel market (2025)** $\times$ (% blended-stretch SKUs) $\times$ (% elastane mass) to compute "poisoned" mass per year; multiply by avoided landfill tipping fee + virgin-fiber replacement cost. (Substack)

Numbers per slide: **maximum 3**. One on the problem (11.3M tons), one on the solution (your benchmark pLDDT/ $\Delta\Delta G$ or % mass recovered), one on the market.

3A. The Hostile Question Bank (and the answer to each)

Question	Your answer structure
"Has any of this been validated in a lab?"	"No. This is a 72-hour computational study at TRL 2-3. The enzyme designs are scored against AlphaFold pLDDT and a published urethanase benchmark. Our 90-day milestone is wet-lab expression at [partner lab]. We do not claim biological validation today."
"What's your TRL?"	"TRL 2-3 in-silico. We have a credible path to TRL 4 — proof of concept in a relevant environment — within 6 months at a partner-lab budget of ~\$45K."
"Why won't this fail like other enzymatic-recycling startups?"	"Two reasons. First, we're not retrofitting an enzyme to an existing fiber — we co-designed both, which removes the kinetic bottleneck that killed earlier attempts. Second, the field has matured: Carbios's planned 50,000 t/yr Longlaville facility (Just Style) (groundbreaking April 2024; (Just Style) commissioning targeted H1 2028, (Intelligent Living) pending financing — Carbios 2024 Universal Registration Document, April 2025) plus 2025 metagenomic urethanase discoveries (PMC12421990) show industrial enzymatic recycling is now investable; we're attacking the elastane-specific gap."
"What does this cost?"	"At pilot scale, modeled \$/ton of polymer recovered: [YOUR VALUE]. For context, Cox's portfolio company Nexus Circular has recycled >30 million pounds of plastic (PR Newswire) (Cox Enterprises, 21 May 2026); our target is to be within 1.5× their unit cost by year 3."
"Who's your customer and will they pay?"	"Two-sided: brands like Carter's, who need to comply with the EU Digital Product Passport (delegated acts 2027; 18-month transition; implementation from 2028 — Intertek, 28 May 2025) and want to extend KIDCYCLE; and recycling operators who need fiber-grade output. Our wedge customer is Carter's Little Planet line."
"What's novel here vs. PHL7, Carbios LCC, UMG-SP-2?"	"Those enzymes target PET or polyurethane films. Ours is a co-designed pair — a new elastane variant + a new urethanase optimized for it. The novelty is the matched-pair design loop, not either component alone."
"Isn't this just Carbios for elastane?"	"Carbios depolymerizes existing PET. We're designing a new fiber that is manufactured to be recyclable from day one with a matched enzyme. It's the difference between catching plastic in a river and building a river that doesn't accept plastic."
"What are the toxic byproducts?"	"Modeled depolymerization products are MDA-free diols and a non-toxic urea derivative — designed in. We pass the predicted SVHC screen in our software. Wet-lab confirmation is gated to Phase 2."
"How is this different from	"Mechanical recycling can't handle blends; chemical recycling can't handle elastane. We're the only proposal that touches both the molecule and the enzyme."

Question	Your answer structure
existing recycling?"	
"What's your moat?"	<i>"Two layers: (1) a co-design dataset linking fiber sequences to enzyme designs that compounds with every wet-lab cycle; (2) the Digital Recyclability Passport as the integration layer with brand ERPs and EU DPP compliance."</i>
"How do you scale this?"	<i>"License the fiber to existing spandex producers (Hyosung, INVISTA); license the enzyme to existing molecular-recycling operators. We are a design-and-IP company, not a manufacturer."</i>

3B. Building Defenses INTO the Demo Itself

The most elegant judge-defense is the demo answering the question before the judge asks it. Concrete UI elements to add:

- **A confidence badge on every prediction:** "pLDDT [YOUR VALUE] · TRL 2 · Not wet-lab-validated."
- **A "what we'd test next" panel** permanently visible in the corner — three bullet milestones with cost and timeline.
- **A safety/toxicity tab** showing predicted byproducts and SVHC screen results, labeled "modeled."
- **A diff view** comparing REWEAR-FUSED's enzyme against UMG-SP-2 baseline, so the "what's novel" question is answered visually.
- **A unit-economics calculator slider** the judge can drag.

When a judge sees these elements **before** they ask, they downgrade three hostile questions into one collegial question.

3C. The "I Don't Know" Structure — Named: Bridge-Hook-Flag

The Grossman Group (David Grossman, yourthoughtpartner.com, "3 Techniques to Master Tough Q&A") gives the verbatim structure: *"Bridging is a great tool to use when you don't have the answer or want to redirect the conversation to a topic that demonstrates your knowledge and credibility... 'I don't know...' (the answer), 'but what I do know...' (the bridge)."* (The Grossman Group)

The complementary Aker Ink PR **Block-and-Bridge** technique: *"Rather than respond with an 'I don't know' or 'no comment' — both of which make you come off as less than open or honest — enlist the block and bridge technique to redirect the interview to something you do know how to*

answer: *'I can put you in touch with someone who knows more about that, but what I can tell you right now is...'*" (Aker Ink)

Full template:

1. **Acknowledge** (1 sec): *"Great question."* (Don't overuse — once per Q&A max.)
2. **Honest concession** (3 sec): *"We haven't tested that specifically."*
3. **Bridge** (3 sec): *"What we **do** know is..."*
4. **Forward commitment** (3 sec): *"...and the test we'd run to nail that down is X, costing Y, in Z weeks."*

Tran VC's deep-tech pitch coaching (tran.vc, 2024): *"Rather than saying 'This part isn't finished yet,' you might say, 'We've validated the first phase, and we're now designing the pilot to test these assumptions further.'"* (Tran vc)

Handling a wrong/hostile judge: never contradict directly. Use the **"and-yes" judo move**: *"Yes — and the way we'd think about that is..."* If the judge is factually wrong, defer to data: *"Our reading of [paper, 2025] is X — happy to share the citation after."* Never say "you're wrong."

3D. Handling Adversarial Technical Depth While Keeping Non-Experts Engaged

When an expert judge drills into the enzyme docking score, **answer them precisely in 1 sentence, then immediately reframe for the room**: *"The $\Delta\Delta G$ is [YOUR VALUE] kcal/mol vs. native — for the non-biochemists, that means we predict the enzyme grabs the fiber roughly $N\times$ more tightly than the wild-type, which is what we need for industrial throughput."* Two sentences. Both audiences served.

3E. Body Language, Voice, Nerves

- **Eye contact:** The 3-Person Rule (Mary Beth Hazeldine, winningpresentations.com): hold one judge's eyes for **one complete thought (3-5 seconds)**, then move. Start with the most-nodding judge to build confidence. (Winningpresentations)
- **Posture:** Feet shoulder-width, weight evenly distributed, hands at waist height in a loose "home position" when not gesturing (Benjamin Ball Associates, 2025). (Benjamin Ball)
- **Voice:** Vary rate and pitch; slow down 30% before each killer number; **pause for 1.5 seconds before saying the stat**. Pauses are powerful (KPMG UK pitch guide: *"pauses are powerful. They make the audience think, and you can catch your breath!"*). (KPMG)
- **Smile** in the opening 5 seconds — it relaxes you and the room (Lumen Learning Public Speaking). (Lumen Learning)
- **Nerves:** physical reset — 4-7-8 breathing for 90 seconds before going on. Adopt a private "power pose" (feet wide, hands on hips) for 2 minutes — the pre-stage ritual itself reduces

anxiety.

- **Mistakes:** never grimace. *"Take a breath and keep going"* (Lumen Learning). The audience doesn't know your script. (Lumen Learning)

PART 4 — TEAM CHOREOGRAPHY (3-Person Team)

4A. Role Division

Role	Owns	During pitch	During Q&A
Lead Presenter (CEO voice)	Pitch narrative, opening, close, business case	Speaks 100% of the stage minutes	Fields business/market/strategy/team questions
Demo Driver	Clicks, screens, demo machine	Silent operation; one rehearsed handoff line at second 1:30 ("Let me show you")	Fields product/UX/demo-specific questions
Science Lead	Enzyme/fiber/biology, TRL framing	Silent unless explicitly handed off to	Fields all technical, validation, mechanism, novelty questions

Single-presenter vs. multi-presenter. Techstars's Nicole Glaros advises "the CEO should give the pitch." Keen.io's Ryan Spahn (TechStars Boulder) describes how their team explicitly broke that advice and gave the pitch to a non-CEO who had the time and presence. **The decision rule:** best presenter, period — not the title. For a 4-minute pitch, single presenter wins (smoother, faster, no hand-off risk). For a 7-minute pitch with deep tech, a single hand-off to the Science Lead at minute 3 for 45 seconds of TRL framing is acceptable. Never hand off more than once.

(Keen)

4B. Rehearsal Plan (the Murder Board)

Murder Board (formalized by NASA's Hans Mark in the 1970s; documented on Wikipedia: Murder board; used today for Supreme Court nominees, cabinet confirmations, and YC Demo Day prep). The board's literal goal is to **kill the proposal** — what survives is what survives the real panel. From Audacia Strategies (2018, 2019): the technique forces founders to face the toughest questions for the first time in the rehearsal room, not "in front of an investor, a client, or a camera." (Wikipedia + 2)

Concrete schedule for the 72-hour build:

- **Hour 48 (24h before pitch):** First end-to-end run-through with the team only. Time it. Trim 30%.
- **Hour 56:** Murder Board #1 — invite 3-5 strangers (other hackathon teams, mentors). Each is assigned a persona: "Cox cleantech exec," "Carter's compliance lead," "skeptical PhD biochemist." Each asks 3-deep follow-ups on every claim (Hunt Big Sales: *"On the key areas, Murder Board members need to push 3 questions deep on that area. The follow up questions are more important than the initial questions."*). [Hunt Big Sales](#)
- **Hour 64:** Rebuild based on feedback. Re-time.
- **Hour 68:** Murder Board #2 — fresh strangers. Record video. Watch back.
- **Hour 70:** Final dress rehearsal in the actual venue (AV check, projector, mic).
- **Hour 71:** Rest. Brad Feld's TechStars NYC chapter ("Practice Like You Play") describes pitches transformed by **6+ full run-throughs** in the final week. Ryan Spahn (Keen): *"I'd follow up a few hour long practice session with a solid hour long nap."* [Keen](#)

Wesley Eames (Techstars): *"We practiced our pitch on everyone... anytime we saw an opportunity to get out of our comfort zone, practice our pitch, and get feedback, we took advantage."* Ryan Spahn: *"I had to practice for at least four hours per day outloud — basically until my voice hurt."* [LinkedIn](#) [Keen](#)

4C. Logistics & Contingency Checklist

The Go-Bag (one team member carries):

- Primary demo laptop (charged, demo mode tested, MP4 on disk)
- Secondary demo laptop (mirrored)
- 4 adapters (USB-C→HDMI, USB-C→VGA, MiniDP→HDMI, lightning→HDMI)
- 2 m HDMI cable + 2 m USB-C cable
- USB-C stick with: MP4 backup, PDF of deck, printable PDF of one-pager
- 20,000 mAh USB-C power bank
- Both laptops' power adapters
- 100 printed one-pagers
- Printed QR-code cards (10× business-card size, 1× A5)
- The Carter's onesie + the petri dish prop
- Wireless presenter clicker (with extra batteries)
- Sharpie, pen, 3-ring sticky-notes
- Bottled water (3)

Pre-demo 90-minute checklist:

- ☐ Both laptops in airplane mode
 - ☐ Both laptops at >95% battery
 - ☐ DND on
 - ☐ Demo MP4 plays from disk on both laptops
 - ☐ Browser at 150% zoom, F11 full-screen
 - ☐ All fonts loaded (visual check)
 - ☐ AV check completed with venue staff
 - ☐ Mic check; volume tested
 - ☐ QR codes scanned successfully by 3 phones at venue lighting
 - ☐ One-pagers stacked at booth
 - ☐ Team eaten, hydrated, used restroom
 - ☐ Opening + closing line memorized verbatim by Lead Presenter
-

PART 5 — WHY STRONG TECHNICAL PROJECTS LOSE, AND THE WINNING DIFFERENTIATORS

5A. Documented Loser Patterns

Synthesized from BizThon ("Top 5 Mistakes Developers Make at Hackathons"), DEV/[code42cate](#) ("This is Why You Are Losing Hackathons"), AngelHack 2026 guide, DoraHacks judging guide, Devpost criteria guide:

1. **Unclear problem framing** ("We built a Next.js app..." instead of "We help overworked nurses with...").
2. **No business case** — pure tech demo with no customer, no \$, no market.
3. **Demo failure** — Wi-Fi, API, GPU crashed.
4. **Over-claiming / over-scoping** — promising commercial-ready in 72 hours.
5. **Ignoring sponsor's actual judging criteria** — [code42cate](#): "If they give you five judging criteria, make it super easy for them to judge." [DEV Community](#)
6. **Feature bloat** — five half-finished features instead of one polished flow.
7. **Burying the lede** — saving the killer stat for slide 8. YC's Justin Kan: "Don't bury the lead." [Y Combinator](#)
8. **Weak delivery** — monotone, no eye contact, reading slides.
9. **Not answering "so what"** — failing to translate the tech to a human outcome.
10. **No fallback for demo failure** — no MP4, no backup laptop.

5B. Winning Differentiators

YC, Techstars, KPMG, Tran VC, and SeedLegals converge on the same patterns:

- **Clarity beats complexity.** YC Michael Seibel: "80% accurate, 100% understandable beats the reverse." (Substack)
- **Lead with strength.** Whatever is most impressive (the molecular animation, the killer stat, the Carter's-aligned KPI) goes first.
- **Show traction or momentum**, even at 72 hours: "We benchmarked against 4 published enzymes in 48 hours."
- **Make the investor talk** (Seibel's "the more they talk, the more they invest").
- **Memorable close.**
- **Calm under pressure** when the demo glitches (SeedLegals, 2025: "How you react under pressure tells them everything about your leadership style"). (SeedLegals)

5C. Sustainability/Cleantech-Specific Judging

Cox judges have read real LCAs. Cox Cleantech's portfolio includes Nexus Circular (advanced recycling, >30M lbs plastic recycled per Cox, 21 May 2026); the Cox Cleantech Accelerator now invests **\$250K per startup** in its 12-week program (gener8tor.com program description; Cox press release, 26 March 2026), with a separate equity-free **Cox Cleantech Residency** providing \$10K stipends and 6 weeks of programming (Cox press release, 14 May 2026). They will detect greenwashing in under 30 seconds. (PR Newswire + 3)

Speak in the cleantech KPI vocabulary (WePitched 2026, Cleantech Group ESG guide, Frontiers in Sustainability 2025):

- **\$ per ton CO₂ avoided** — target sub-\$100/ton at scale. (WePitched)
- **Scope 4 emissions** — emissions your product **prevents** in the wider world (vs. Scope 1-3 which measure your own footprint). The 2026 cleantech-investor "North Star" metric per WePitched. (WePitched)
- **Additionality** — prove your impact wouldn't happen anyway through existing market/regulatory shifts. WePitched (2026): the most common founder mistake is "claiming impact that would have happened anyway." (WePitched) (WePitched)
- **Circular Material Use Rate (CMUR), Product Circularity Index (PCI), and Lifecycle Carbon Intensity (LCI)** — Frontiers in Sustainability (2025) documents up to "30% reductions in lifecycle carbon intensity" from CE-practicing firms. (Frontiers)

For REWEAR-FUSED, the headline cleantech claim should be expressed as: "If 10% of U.S. blended-elastane children's apparel converted to REWEAR-FUSED, we'd unlock ~\[YOUR

VALUE\] tons/year of recoverable polymer from the existing landfill stream — at an additional cost of <\[YOUR VALUE\] per garment, putting us at ~ \[YOUR VALUE\]/ton CO₂ avoided at scale."* Always with a transparent bottom-up calculation visible to the judge.

Greenwashing red flags to actively avoid: vague "green," "eco-friendly," "sustainable" without quantified delta. Cite real circular-economy metrics, not adjectives.

5D. The Closing Line — Memorable Closes

Memorable closes are **specific, sensory, and callback the opening**. Bad: *"Thank you for your time."* Good: *"A 4-year-old will outgrow this onesie in 90 days. We want the next one to outlive the kid — in a closed loop, and from Atlanta."*

The closing line must:

1. Echo the opening image (the onesie).
 2. Compress the entire vision into ≤2 sentences.
 3. Plant a flag the judge can quote back to a colleague after lunch.
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Recommendations

THE DEFINITIVE 72-HOUR PLAYBOOK (in chronological order)

Hours 0–4: Lock the lede.

- Choose one killer stat (candidate: EPA 11.3M tons U.S. textile MSW landfilled 2018; Carter's 19% U.S. baby apparel 0–2 per Statista/Carter's investor data 2022). (Statista)
- Write the opening sentence and the closing sentence first. Memorize both.
- Identify TRL as 2–3 and write the verbatim TRL slide.

Hours 4–24: Build the deterministic demo.

- Static build, localhost only. All API calls mocked.
- One scripted path, no branches, no live inference.
- Pre-render the 3D molecular animation as MP4; use 3Dmol.js or Mol* for the interactive overlay.
- Self-host fonts.

Hours 24–40: Build the defense UI.

- Add the confidence badge ("pLDDT [YOUR VALUE] · TRL 2 · Not wet-lab-validated") to every prediction screen.
- Add the "what we'd test next" sidebar.
- Add the predicted-byproducts safety panel.
- Add the LCA / Scope 4 / \$-per-ton-CO₂ calculator slider, bottom-up math visible.

Hours 40–48: Record the backup MP4.

- 90 seconds, 1080p, H.264, captions burned in, narrated.
- Save in 3 locations: laptop A, laptop B, USB stick.
- Rehearse the verbatim pivot line: *"Let me show you the recording from this morning."*

Hours 48–56: First Murder Board.

- 3–5 strangers. Personas: Cox cleantech exec, Carter's compliance lead, skeptical biochemist.
- 3-deep follow-ups on every claim.
- Record. Watch back. List every weakness.

Hours 56–64: Rebuild + business case.

- Tighten the answer bank for the 11 hostile questions in Part 3A.
- Finalize the bottom-up market sizing slide (anchor on Statista's \$54.62B U.S. childrenswear 2025).
- Tie every line to a Carter's-internal metric (KIDCYCLE relaunch 2025, OEKO-TEX 99% 2025, EU DPP 2027/2028).

Hours 64–70: Second Murder Board + final dress rehearsal.

- Fresh strangers. Time exactly to 95% of allotted slot.
- Venue dress rehearsal with the real AV.

Hour 70–72: Rest, hydrate, eat. Sleep is part of the playbook.

Stage-gate thresholds (what changes the plan)

- If the murder board cannot understand the loop in 30 seconds → **simplify the diagram** before doing anything else.
- If demo MP4 isn't done by hour 48 → **stop building features**; the MP4 outranks every feature.

- If the killer stat is challenged as inaccurate → **swap it** to the next-most-cited source; never defend a wobbly number on stage.
- If the team is exhausted at hour 60 → cut a feature, not a rehearsal. Rehearsal is the highest-ROI hour.

The Anticipated Q&A Bank — REWEAR-FUSED-tailored

(See Part 3A table above for the 11 most likely hostile questions and the verbatim structured answers. Print this as the back of each team member's one-pager and laminate.)

The Rehearsal & Contingency Plan

- **Minimum 6 full end-to-end run-throughs** in the final 48 hours.
 - **Minimum 2 murder boards** with external strangers, each with 3-deep follow-ups.
 - **One full venue dress rehearsal** with real AV.
 - **Demo fallback hierarchy** rehearsed verbatim: interactive WebGL → static high-res renders → recorded flythrough → 90-second narrated MP4.
 - **Mirrored backup laptop**, cabled and warm, on the table the entire pitch.
 - **Pre-pitch DND/airplane-mode/sleep-disabled checklist** taped to the laptop lid.
-

Caveats

- **Event format is not fully verifiable publicly.** Cox's publicly described "Play With Purpose" is a community/volunteerism platform tied to its role as a FIFA World Cup 26™ Atlanta Host City Supporter (coxenterprises.com/our-impact/fifa-world-cup-2026-atlanta, 2025–2026), and no public source we could locate confirms a Cox hackathon by that exact name, a Carter's "Making and Remaking" track, or a FIFA World Cup 2026 tickets grand prize through any such hackathon. **Request the official sponsor brief / judging rubric from the organizers** and re-weight the playbook against the actual published criteria. Judging-criteria specificity matters (DoraHacks, 2026: "*Innovation: Does this project demonstrate an innovative integration with our API?*" is evaluable; "*Is this project exciting?*" invites debate) — get the rubric in writing. (DoraHacks)
- **The Carbios Longlaville plant is NOT yet operational.** Per Carbios's own 2024 Universal Registration Document (filed April 30, 2025) and H1 2025 results: the April 2024 groundbreaking was followed by a 6–9 month construction deferral in December 2024 due to funding delays; **commissioning is now targeted for H1 2028, conditional on finalizing financing.** Do not call it a "success." Frame it as: "*Carbios's planned 50,000 t/yr Longlaville facility (groundbreaking April 2024; commissioning now targeted H1 2028, pending financing).*" (Just Style) (Intelligent Living)

- **Carter's market share is 19%, not "~20%, Trellis 2025."** Per Statista citing Carter's own investor filings (2022): *"Carter's brand held a 19 percent share of the baby apparel market for ages zero to two."* Trellis is also the name of a Carter's in-house organic-cotton sub-brand (Carter's Little Planet by Trellis); do not conflate the two. (Statista)
- **U.S. childrenswear market size.** Per Statista's U.S. Children's Apparel Market Outlook (2025): *"In 2025, the revenue in the Children's Apparel market in the United States amounts to US\$54.62bn."* This covers ages 0-14; do bottom-up math to size the wedge segment (blended-stretch baby apparel) precisely. (Statista)
- **Cox Cleantech program structure.** The **\$250K per company** investment is the **12-week Cox Cleantech Accelerator** (gener8tor.com; Cox press release 26 March 2026), and is distinct from the **6-week equity-free Cox Cleantech Residency** announced 14 May 2026 with a \$10K stipend. Do not conflate the two programs or their dates. (gener8tor + 3)
- **EU Digital Product Passport timing.** Use the verifiable framing: *"textile-specific delegated acts expected to be adopted in 2027, followed by a minimum 18-month transition period, meaning implementation is expected to start from 2028 onward"* (Carbonfact, citing EC ESPR Working Plan COM(2025)187 final; Intertek, 28 May 2025). Don't tighten this to a single year. (Carbonfact)
- **The TRL self-assessment depends on the team's actual benchmark scores.** All pLDDT and $\Delta\Delta G$ numbers in this playbook are placeholders ([YOUR VALUE]); substitute your real benchmark values and never invent them.
- **a16z Bio's 2018 "16 Pitfalls" guidance** was framed for computational therapeutics, not materials/textiles, but its core warning (in-silico must be backed by wet-lab) applies directly. Use the verbatim Pande/Conde quote rather than paraphrasing.